

Shakir Hossain . Jarvis Consulting

An experienced Data Engineer with over 3 years of hands-on experience designing and implementing cloud data pipelines, database solutions, and large-scale migrations on Azure and modern data platforms. A proven track record collaborating with cross-functional teams and asset management stakeholders to deliver reliable, business-critical data products.

Skills

Proficient: Python, Linux/Bash, RDBMS/SQL, PostgreSQL, Agile/Scrum, Git, Docker

Competent: Java, PySpark, Pandas, Dbt, Airflow, Scala, YAML, Tableau, PowerBi, Kafka

Familiar: Azure DevOps, Azure Data Factory, Snowflake, Databricks, Hadoop, Ubuntu, MSSQL, AWS

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_ShakirHossain

Cluster Monitor [GitHub]: The Jarvis LCA team needed better visibility into cluster resource utilization to plan capacity and to mitigate performance issues. Designed and implemented a Bash-based monitoring agent, scheduled via cron, to collect real-time node usage metrics and push them into a central store for analysis. Enabled quantitative insights into usage patterns, supporting more accurate capacity planning and reducing the risk of over/under-provisioning. This deepened my understanding of lightweight monitoring on Linux, cron-based scheduling, and how small automation scripts can materially improve operational decision-making.

Python Data Analytics [GitHub]: LGS, an online retail platform, relied on generic marketing due to limited insights from raw transactional data. Analyzed and transformed transactional data to identify customer purchasing patterns and behavioral trends. Enabled data-driven personalization, improving the platform's ability to target customers with more relevant marketing strategies. Learned how structured data analysis can convert raw data into actionable insights that directly influence business strategy and customer engagement.

Databricks Analytics Project [GitHub]: The existing data solution struggled to scale efficiently with growing data volumes and processing demands. Redesigned the data architecture using Apache Spark within a Databricks workspace to enable distributed, multi-node data processing. Achieved scalable big data processing with improved performance, reliability, and faster data pipelines. Gained hands-on experience in designing and operating distributed data solutions, and learned how cluster-based processing significantly improves scalability and throughput.

Jarvis Hive Project [GitHub]: The organization needed to analyze large-scale datasets stored in Google Cloud Storage using a Hadoop-based data platform, but query performance was a challenge. Exported data from Google Cloud Storage to a Hadoop cluster, wrote and optimized Hive queries with partitioning, and applied columnar file optimizations. Improved query performance and reduced runtime execution, enabling faster and more efficient data analysis. Learned how data movement, Hive partitioning, and columnar storage formats work together to optimize performance in large-scale data processing systems.

Highlighted Projects

Brave Career Data Engineering Bootcamp [GitHub]: Designed and Implemented data pipelines using dbt Core to aggregate and store key metrics for an e-commerce platform. Monitored inventory levels, stock turnover, and supply chain efficiency to ensure optimal stock levels and minimize costs.

Machine Learning Project with Alexey Grigorev [GitHub]: Generated ML models and evaluated based on Classification and Regression Metrics. Transformed ML applications into serverless web services using Docker, Kubernetes and AWS Lambda.

Professional Experiences

Data Engineer, Jarvis (2025-present): Complex technical data and system behaviors needed to be clearly communicated and validated across teams. Created data visualizations to synthesize technical information, wrote scripts for stored procedures and triggers, and documented results of software application, information, and telecommunication system tests. Improved clarity of technical insights, ensured reliable database operations, and provided thorough documentation to support system quality and maintenance. Strengthened skills in data visualization, database programming, and technical documentation, and learned the importance of clear communication in system validation and decision-making.

Solutions Developer, Emera Energy (2021-2025): The organization relied on legacy on-premise SQL Server infrastructure running on Windows Server 2008 R2, leading to high maintenance overhead and scalability limitations. Designed and executed MSSQL and MySQL migrations from on-premise environments to Azure SQL Services, and built end-to-end proof-of-concept ETL pipelines using Azure Data Factory to validate Azure PaaS solutions against legacy tools like SSIS and linked servers. Successfully decommissioned outdated infrastructure, reduced maintenance costs, and demonstrated the viability and benefits of Azure PaaS-based data integration. Gained deep experience in cloud database migrations and learned how modern PaaS ETL tools can simplify architecture, improve scalability, and replace legacy frameworks effectively.

Software Developer, LED Roadway LTD (2019-2020): The company needed an accessible way to monitor the geo-location and performance of streetlamp microsensors in real time. Developed an Android application that integrated sensor performance and location data, and built a geographical user interface using the OpenStreetMap API to display live data feeds and sensor locations. Improved accessibility and visibility of sensor data, enabling users to monitor streetlamp performance and locations efficiently through a mobile interface. Gained experience in mobile app development, real-time data integration, and building map-based user interfaces to enhance usability and operational monitoring.

Education

Dalhousie university (2018-2021), Bachelor's in Computer Science, Faculty of Computer Science

Saint Mary's university (2016-2018), Diploma of Electrical Engineering, Faculty of Engineering

Miscellaneous

- Microsoft Azure Certified- Azure Database Administrator Associate DP-300
- Chess Enthusiast
- Recreational Golfer